Probability Theory and Statistics, Tutorial 4

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Requirements for a Binomial Experiment

Key Conditions

1. Fixed number of trials (n)
   - Sequence of n smaller experiments
   - n is fixed in advance
2. Dichotomous trials
   - Two possible outcomes: Success (S) or Failure (F)
   - Labels S and F are arbitrary
3. Independent trials
   - Outcome of one trial doesn't affect others
4. Constant probability
   - P(S) = p remains constant across trials

For $X \sim \text{Bin}(n, p)$, the pmf is:

$$b(x; n, p) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x} & x = 0, 1, 2, ..., n \\ 0 & \text{otherwise} \end{cases}$$

$$B(x; n, p) = P(X \leq x) = \sum_{y=0}^{x} b(y; n, p)$$

Important Properties

For $X \sim Bin(n, p)$:

$$E(X) = np$$
$$V(X) = np(1-p) = npq$$
$$\sigma_X = \sqrt{npq}$$

where $q = 1 - p$

Table A.1 Cumulative Binomial Probabilities *(cont.)*

$$B(x; n, p) = \sum_{y=0}^{x} b(y; n, p)$$

e. $n = 25$

$$B(x; n, p) = P(X \le x) = \sum_{y=0}^{x} b(y; n, p)$$

		p														
		0.01	**0.05**	**0.10**	**0.20**	**0.25**	**0.30**	**0.40**	**0.50**	**0.60**	**0.70**	**0.75**	**0.80**	**0.90**	**0.95**	**0.99**
	0	.778	.277	.072	.004	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
	1	.974	.642	.271	.027	.007	.002	.000	.000	.000	.000	.000	.000	.000	.000	.000
	2	.998	.873	.537	.098	.032	.009	.000	.000	.000	.000	.000	.000	.000	.000	.000
	3	1.000	.966	.764	.234	.096	.033	.002	.000	.000	.000	.000	.000	.000	.000	.000
	4	1.000	.993	.902	.421	.214	.090	.009	.000	.000	.000	.000	.000	.000	.000	.000
	5	1.000	.999	.967	.617	.378	.193	.029	.002	.000	.000	.000	.000	.000	.000	.000
	6	1.000	1.000	.991	.780	.561	.341	.074	.007	.000	.000	.000	.000	.000	.000	.000
	7	1.000	1.000	.998	.891	.727	.512	.154	.022	.001	.000	.000	.000	.000	.000	.000
	8	1.000	1.000	1.000	.953	.851	.677	.274	.054	.004	.000	.000	.000	.000	.000	.000
	9	1.000	1.000	1.000	.983	.929	.811	.425	.115	.013	.000	.000	.000	.000	.000	.000
	10	1.000	1.000	1.000	.994	.970	.902	.586	.212	.034	.002	.000	.000	.000	.000	.000
	11	1.000	1.000	1.000	.998	.980	.956	.732	.345	.078	.006	.001	.000	.000	.000	.000
x	**12**	1.000	1.000	1.000	1.000	.997	.983	.846	.500	.154	.017	.003	.000	.000	.000	.000
	13	1.000	1.000	1.000	1.000	.999	.994	.922	.655	.268	.044	.020	.002	.000	.000	.000
	14	1.000	1.000	1.000	1.000	1.000	.998	.966	.788	.414	.098	.030	.006	.000	.000	.000

Exercise 1

NBC news reported that 1 in 20 children in the US has a food allergy. Consider selecting a random sample of 25 children and let X be the number of children in the sample who have a food allergy.

(a) What distribution does X follow?

(b) Calculate $P(X \leqslant 3)$, $P(X < 3)$, $P(X = 3)$, $P(X \geqslant 4)$, and $P(1 \leqslant X \leqslant 2)$.

(c) Determine $\mathbf{E}(X)$ and σ_X.

(d) In a random sample of 50 children, what is the probability that none has a food allergy?

Exercise 2

A toll bridge charges $1 for passenger cars and $2.5 for other vehicles. Suppose that during daytime hours, 60% of all vehicles are passenger cars. If 25 vehicles cross the bridge during a particular daytime period, what is the resulting expected toll revenue?

In a **Poisson process**, $\{N(t)\}_{t\geq 0}$ is a *counting process*:

- $N(t)$ is a **random variable** (depends on the randomness of event times).
- It counts **how many events have occurred from time 0 up to time t**.
- So $N(t) \in \{0, 1, 2, \dots\}$, it is nondecreasing in t, and $N(0) = 0$.

Example: if events are "arrivals to a queue", then $N(5) = 3$ means **3 arrivals occurred by time 5**.

What is X in the Poisson distribution definition?

In pic 2, X is just a **generic discrete random variable** with a Poisson distribution:

$$\mathbb{P}(X = x) = p(x; \mu) = \frac{e^{-\mu}\mu^x}{x!}, \quad x = 0, 1, 2, \dots$$

Here $\mu > 0$ is the Poisson mean parameter.

So X is *not automatically a process*—it's one Poisson-distributed random variable.

How are $N(t)$ and X connected?

For a Poisson process with rate α, **for each fixed time** t the count $N(t)$ has a Poisson distribution with parameter $\mu = \alpha t$. That is:

$$N(t) \sim \text{Poisson}(\alpha t).$$

So if you want to match the notation in the Poisson pmf definition, you can literally set

$$X := N(t), \qquad \mu := \alpha t.$$

Then the pmf becomes

$$\mathbb{P}(N(t) = x) = \mathbb{P}(X = x) = \frac{e^{-\alpha t}(\alpha t)^x}{x!}, \quad x = 0, 1, 2, \dots$$

Definition

A discrete random variable X has a Poisson distribution with parameter μ ($\mu > 0$) if:

$$p(x;\mu) = \frac{e^{-\mu}\mu^x}{x!}, \quad x = 0, 1, 2, 3, \ldots$$

where:

- $e \approx 2.71828$ (base of natural logarithm)

Key Result

For $X \sim \text{Poisson}(\mu)$:

- $E(X) = \mu$
- $V(X) = \mu$
- $\sigma = \sqrt{\mu}$

Poisson process (rate α) — properties

A counting process $\{N(t)\}_{t\geq 0}$ is a **Poisson process with rate** $\alpha > 0$ if it satisfies:

1. **Independence (independent increments).**
   For disjoint time intervals, the numbers of events are independent.
2. **Stationary increments.** (memoryless)
   For $s, t \geq 0$, the distribution of the increment $N(t+s) - N(s)$ depends only on the length t, not on s.
3. **Poisson distribution of counts.**
   For each $t \geq 0$,

   $$N(t) \sim \text{Poisson}(\mu), \qquad \mu = \alpha t,$$

   i.e. for $x = 0, 1, 2, \ldots$,

   $$\mathbb{P}(N(t) = x) = p(x;\mu) = p(x;\alpha t) = \frac{e^{-\alpha t}(\alpha t)^x}{x!}.$$

   Here α is the **average rate (events per unit time)**.
4. **No simultaneous events.**
   The probability that two or more events occur at exactly the same instant is 0.
   (Equivalently, over a very small interval of length Δt, the chance of ≥ 2 events is negligible.)

Exercise 3

The number of requests for assistance received by a towing service is a Poisson process with rate $\alpha = 4$ per hour.

(a) Compute the probability that exactly 10 requests are received during a particular 2-hour period.

(b) If the operators of the towing service takes a 30-minute break for lunch, what is the probability that they do not miss any calls for assistance during their break?

(c) How many calls would you expect during their break?

(d) Let X denote the number of requests that will be received during the lunch break. Find the standard deviation of X.

(e) Suppose that t minutes ($0 < t < 30$) after the start of the lunch break, an operator goes back to the office and finds that no request has been received during the first t minutes. Given this, what is the probability that no request will be received during the break?

Here is one solution for Exercise 3 (e)

We are to find the conditional probability $P(X_{0.5} = 0 \mid X_{t/60} = 0)$, which, by definition, equals

$$\frac{P(\{X_{0.5} = 0\}) \cap \{X_{t/60} = 0\})}{P(X_{t/60} = 0)}.$$

Note that $\{X_{0.5} = 0\} \subseteq \{X_{t/60} = 0\}$, and so $\{X_{0.5} = 0\} \cap \{X_{t/60} = 0\} = \{X_{0.5} = 0\}$. Since $X_{0.5} \sim \text{Pois}(2)$ and $X_{t/60} \sim \text{Pois}(t/15)$, we have

$$P(X_{0.5} = 0 \mid X_{t/60} = 0) = \frac{P(X_{0.5} = 0)}{P(X_{t/60} = 0)} = \frac{\mathrm{e}^{-2} \cdot \dfrac{2^0}{0!}}{\mathrm{e}^{-t/15} \cdot \dfrac{(t/60)^0}{0!}} = \mathrm{e}^{\frac{t}{15}-2} = \mathrm{e}^{-4\left(0.5-\frac{t}{60}\right)}.$$

Notice that this is the same as the probability of the event that no request is received in $0.5 - \frac{t}{60}$ hour (i.e., the remaining time), which is due to the fact tha Poisson distribution is "memoryless".

Table A.2 Cumulative Poisson Probabilities *(cont.)*

$$F(x; \mu) = \sum_{y=0}^{x} \frac{e^{-\mu}\mu^{y}}{y!}$$

	μ										
x	**2.0**	**3.0**	**4.0**	**5.0**	**6.0**	**7.0**	**8.0**	**9.0**	**10.0**	**15.0**	**20.0**
0	.135	.050	.018	.007	.002	.001	.000	.000	.000	.000	.000
1	.406	.199	.092	.040	.017	.007	.003	.001	.000	.000	.000
2	.677	.423	.238	.125	.062	.030	.014	.006	.003	.000	.000
3	.857	.647	.433	.265	.151	.082	.042	.021	.010	.000	.000
4	.947	.815	.629	.440	.285	.173	.100	.055	.029	.001	.000
5	.983	.916	.785	.616	.446	.301	.191	.116	.067	.003	.000
6	.995	.966	.889	.762	.606	.450	.313	.207	.130	.008	.000
7	.999	.988	.949	.867	.744	.599	.453	.324	.220	.018	.001
8	1.000	.996	.979	.932	.847	.729	.593	.456	.333	.037	.002
9		.999	.992	.968	.916	.830	.717	.587	.458	.070	.005
10		1.000	.997	.986	.957	.901	.816	.706	.583	.118	.011
11			.999	.995	.980	.947	.888	.803	.697	.185	.021
12			1.000	.998	.991	.973	.936	.876	.792	.268	.039
13				.999	.996	.987	.966	.926	.864	.363	.066
14				1.000	.999	.994	.983	.959	.917	.466	.105
15					.999	.998	.992	.978	.951	.568	.157
16					1.000	.999	.996	.989	.973	.664	.221
17						1.000	.998	.995	.986	.749	.297
18							.999	.998	.993	.819	.381
19							1.000	.999	.997	.875	.470
20								1.000	.998	.917	.559

10. Suppose small aircraft arrive at a certain airport according to a Poisson process with rate $\alpha = 8$ per hour, so that the number of arrivals during a time period of t hours is a Poisson random variable with parameter $\mu = 8t$.

(a) What is the probability that exactly 6 small aircraft arrive during a 1-hour period? At least 6? At least 10?

(b) What are the expected value and standard deviation of the number of small aircraft that arrive during a 90-minute period?

(c) What is the probability that at least 20 small aircraft arrive during a 2.5-hour period? That at most 10 arrive during this period?

Continuous Random Variable

Two key properties:

- Possible values comprise either:
  - A single interval on the number line, OR
  - A union of disjoint intervals
- For any number c, $P(X = c) = 0$

Definition

Let X be a continuous rv. A probability density function (pdf) of X is a function $f(x)$ such that for any two numbers a and b with $a \leq b$:

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

Key Property

If X is a continuous rv with pdf $f(x)$ and cdf $F(x)$, then at every x where the derivative exists:

$$F'(x) = f(x)$$

Definition

The expected or mean value of a continuous rv X with pdf $f(x)$ is:

$$\mu_X = E(X) = \int_{-\infty}^{\infty} x \cdot f(x)dx$$

General Property

If X is continuous with pdf $f(x)$ and $h(X)$ is any function of X:

$$E[h(X)] = \mu_{h(X)} = \int_{-\infty}^{\infty} h(x) \cdot f(x)dx$$

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2$$

Definition

The cumulative distribution function $F(x)$ for a continuous rv X is defined for every number x by:

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(y)dy$$

The probability mass function (pmf) of a discrete rv is defined for every number x by:

$$p(x) = P(X = x) = P(\text{all } \omega \in S : X(\omega) = x)$$

Definition

A continuous rv X is said to have a uniform distribution on the interval $[A, B]$ if the pdf of X is:

$$f(x; A, B) = \begin{cases} \frac{1}{B-A} & A \leq x \leq B \\ 0 & \text{otherwise} \end{cases}$$

Exercise 4

A college professor never finishes his lecture before the end of the hour and always finishes within 2 minutes after the hour. Let X be the time that elapses between the end of the hour and the end of the lecture, which pdf

$$f(x) = \begin{cases} kx^2, & \text{if } 0 \leqslant x \leqslant 2 \\ 0, & \text{otherwise} \end{cases}$$

(a) Identify the value of k, and then derive the cdf for X.

(b) Find the probability that the lecture ends within 1 minute of the end of the hour.

(c) What is the probability that the lecture continues beyond the hour for between 60 and 90 seconds?

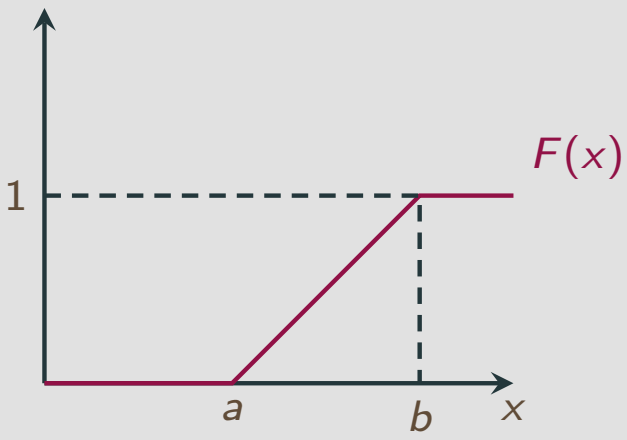
(d) What is the probability that the lecture continues for at least 90 seconds beyond the end of the hour?

(e) If the probability that the lecture ends within m minutes of the end of the hour equals 0.5, find the value of m.

(f) Calculate $\mathbf{E}(X)$ and $V(X)$.

Exercise 5

The graph of the cdf of a random variable X is shown in the figure below. The graph consists of three phases, each being a segment or a ray.



(a) Find the expressions of the cdf and pdf of X, respectively.

(b) Which type of distribution does X follow?

(c) Calculate the expected value and variance of X.